

Paper Directions 12 — Transcript guidance (May 20, 2026)

Filename: 20260520_Transcript_12_guidance.md

Source: transcripts/20260520_Paper_Directions_12_otter_ai_transcript.docx (Otter; ~16 min)

Participants: Alex Gates, Charles Levine

Interpreted in: sports/documents/Alex_model_interpreted.md (PD12 addendum)

Purpose: Standalone summary of what Alex prioritized in PD12 — especially the **four workstreams** — without reading the full transcript or interpreted memo.

Provenance note

- Alex_model.md was produced in ~45 minutes with AI assistance from prompts — a **structured theoretical brief**, not finalized paper prose.
- PD12 overrides or sharpens several terms (especially **capability** vs **crowding/quality**) where the live discussion went deeper than the draft text.

Critical conceptual fix (read before the four priorities)

Capability ≠ organizational strength

Misread	Corrected read (Alex, PD12)
"Capability" = program/institutional strength (e.g. Duke's brand)	"Capability" ≈ individual talent and how differentiated the person is from peers
Elite school = high org quality only	Elite environment = very high talent but extreme substitutability — excellent people, hard to tell apart

Implementation mapping:

Construct	Empirical proxy (MBB)	Role
Quality $\bar{a}_t$	LOO mean teammate performance (<code>poolq_loo</code>)	Rising leg of inverted-U
Congestion $C_{i,t}$	Viable-peer count / <code>crowding_smooth</code> above threshold θ	Downturn leg — do not use LOO sum ($\approx$ mean at fixed roster size)

“How many people are above the threshold—that’s an easy calculation.” — Alex (~7:20)

Charles committed on the call (~8:22) to make **congestion crowding** first-class in the empirical pipeline.

Alex’s four priorities (second half of PD12)

Alex separated **four threads**. They are **not the same task**.

Priority 1 — Parameter identifiability (simulation + fit)

Task: Fit the generative model (~5–6 parameters) to **real data in all three domains** in scope (MBB, tenure, ...).

Alex’s concern: The model may be **overparameterized** — many parameter combinations fit similarly (curve-fitting inverted-U shapes without unique structural meaning).

Remedies discussed:

- **Fix** some generative knobs across worlds (e.g. same Beta hyperparameters everywhere — not estimated per dataset).
- **Roll** mechanism-specific terms into a single outcome Y per domain (“in this dataset we roll a, b, c into Y ”).
- **Lower dimensionality** until recovered parameters are **unique and interpretable**.

Distinction: Identifiability = “Can we estimate the structural parameters of this model from these tables?”

Repo hook: `539_alex_model.ipynb` — fit / identifiability sweeps.

Priority 2 — Extreme events (simulation sensitivity)

Task: Knob-turning in the sim, not rare draft surprises in real data.

Examples:

- Set $\lambda \rightarrow 0$
- Turn congestion off
- Tighten viability thresholds

Goal: Confirm inverted-U and mediation patterns **disappear mechanistically** when they should.

Distinction: Validate **internal logic** of the generative model before over-interpreting one empirical ventile plot.

Repo hook: `539_alex_model.ipynb` — parameter kill switches; 538D CELL 10 ability-only null (talent-only fails).

Priority 3 — Model-guided empirical features

Task: Theory should **propose new measurements**, not only fit old ones.

Flagship example (MBB):

- Quantify explicit **congestion** $C_{i,t}$
- Test whether $C_{i,t}$ **mediates** downturn in $\bar{a}_t$
- Natural experiments: **Kentucky / Duke mega-recruit years**, transfer shocks, mobility reversal (leave congested program → odds rise)

Charles's design constraint (~12:15): Ideal congestion measure **reflects talent in the system indirectly** but **does not enter the outcome as raw** a_i — study “how crowded is this team?” and predict harm to **marginal** members.

Near-threshold population: Inverted-U harm is strongest for **borderline** performers (NIH/Wang “just at the cut line”), not obvious superstars or non-contenders. **Superstars still get drafted** in elite teams — heterogeneous effect, not “elite hurts everyone.”

Repo hook (June 2026): `crowding_smooth` in `tier1_pool_assignment.py` + 538D CELL 10; CELL 4D heterogeneity exports; `tier1_mechanism_vars.py` / 530 export for viable-peer counts.

Priority 4 — Falsification and scope

Task: Strong paper shows **where the model fails**, not only where it works.

Alex's agenda:

- Find a **4th or 5th setting** where inverted-U **does not** appear (NHL draft mentioned) → explain **which assumption breaks** (attention structure, evaluation noise, no local peer congestion for slots, etc.).
- Explore **new positive domains** if data exist (Charles suggested macro: GDP, patents, oil production vs. national firm competition; Alex noted “really nice level of estimation”).

Pace: “Let seep over weeks” — not blocking immediate MBB congestion work.

Six mechanisms — scope discipline

`Alex_model.md` lists six mechanism classes. PD12 walkthrough: they are **reasons the theory matters**, not six separate causal channels to prove in one MBB table.

#	Mechanism	PD12 emphasis
1	Comparative evaluation	Signal distinctiveness in elite groups
2	Attribution dilution	Include in narrative; optional test later
3	Finite attention	Substitutability of borderline elite; near-threshold
4	Organizational risk minimization	"Moneyball" heuristics when $C_{i,t}$ high
5	Opportunity suppression	Recursive — peers → fewer touches → weaker stats
6	Queueing / timing	Selection scarcity; don't over-build all facets in v1

Writing rule: Address **some**, not all; use mechanisms to **guide measurement**.

Army evidence shown (same meeting)

Charles showed Alex **100-bin equal-width CIF** (promotion top, attrition bottom) on LOO pool quality.

PD12 point: Cross-domain shape exists — not NCAA-only curve-fit. Motivated investing in MBB congestion crowding.

Careful read: Promotion and attrition panels are **not** required to mirror bin-for-bin; late-bin attrition rebound ≠ superstar story without checking bin N and uncertainty.

Suggested implementation order (from interpreted memo)

- 1. **530 export:** `peer_viable_count_loo` (+ optional smooth $C_{i,t}$); keep `poolq_loo` as quality leg.
- 2. **538 / 538D CELL 10:** `crowding_smooth` live; document Plot B axis (pool mean vs LOO pool quality).
- 3. **539:** Priorities 1–2 (identifiability, kill switches).
- 4. **combine_bridge / panel:** Optional `Y_nba_signal` for θ robustness.
- 5. **Writing:** PD12 cross-ref in advisor briefs; **do not** claim all six mechanisms separately identified in MBB v1.

PD12-aligned one-paragraph paper sentence

We distinguish **team quality** (mean peer performance, $\bar{a}_t$) from **viable-peer congestion** (density of teammates above a prospect threshold, $C_{i,t}$). Elite environments raise both average talent and **substitutability**; the inverted-U in advancement is predicted to steepen on the right primarily where $C_{i,t}$ is high, especially for **near-threshold** individuals, while **top latent talent** still clears selection—consistent with the small recovery at the highest ventiles in our NCAA draft data.

How PD12 maps to current v1 forward plan (June 2026)

PD12 priority	v1 manuscript status (agent consensus)
1 Identifiability	Deferred — Path II qualitative generative POC; not 3-domain parameter fit
2 Extreme events	Partial — talent-only fails (CELL 10); full kill-switch sweeps post-draft
3 Congestion features	Active — aligns with D10 bundle, near-threshold prediction #1, crowding in score
4 Falsification	Deferred — NHL / 4th domain post-Summer-Fall 2026

PD12 is **richer** than the locked v1 path. The correspondence experiment **consciously parked** 1 and 4 to enable **writing now**; Priority 3 is what SCOUT D10 delivers.

Open questions left on the table (PD12)

- Exact θ for MBB: global PPM quantile vs combine/draft-based $Y_{\text{nba_signal}}$ vs smooth logistic viability (539).
- Mediate with $C_{i,t}$ in LPM only vs also in Tier 1 generative selection (538).
- NHL (or other) **negative case** data feasibility for Priority 4.

Related files

File	Role
<code>Alex_model_interpreted.md</code>	Full PD12 addendum + mechanism notes
<code>Alex_model.md</code>	AI-assisted theoretical brief (pre-PD12 corrections)
<code>Tier1_Presorting_Design_Note.md</code>	PD11 (earlier meeting — pool pre-sorting; separate from PD12 four priorities)
<code>forward_plan_background.md</code>	Bring-up-to-speed after agent experiment
<code>20260615_1045_COMPASS_to_Charles_forward_plan_and_questions.md</code>	Ordered next steps + Charles locks

End PD12 guidance summary.